
Database Tables Design

1 Categories Table

Attribute	Data Type	Constraints
CategoryID	INT	PRIMARY KEY, IDENTITY(1,1)
CategoryName	VARCHAR(100)	NOT NULL, UNIQUE

2 Medicines Table

Attribute	Data Type	Constraints
MedicineID	INT	PRIMARY KEY, IDENTITY(1,1)
Name	VARCHAR(100)	NOT NULL
Description	VARCHAR(255)	NULL
Price	DECIMAL(10,2)	NOT NULL, CHECK (Price > 0)
QuantityInStock	INT	NOT NULL, DEFAULT 0, CHECK (>= 0)
MinimumStockLevel	INT	NOT NULL, CHECK (>= 0)
CategoryID	INT	NOT NULL, FOREIGN KEY

3 Suppliers Table

Attribute	Data Type	Constraints
SupplierID	INT	PRIMARY KEY, IDENTITY(1,1)
Name	VARCHAR(100)	NOT NULL
Phone	VARCHAR(20)	UNIQUE
Address	VARCHAR(255)	NULL

4 SupplierInvoices Table

Attribute	Data Type	Constraints
InvoiceID	INT	PRIMARY KEY, IDENTITY(1,1)
SupplierID	INT	NOT NULL, FOREIGN KEY
InvoiceDate	DATE	NOT NULL, DEFAULT GETDATE()

Attribute	Data Type	Constraints
TotalAmount	DECIMAL(10,2)	CHECK (>= 0)

5 InvoiceDetails Table (حل العلاقة Many-to-Many)

Attribute	Data Type	Constraints
InvoiceDetailID	INT	PRIMARY KEY, IDENTITY(1,1)
InvoiceID	INT	NOT NULL, FOREIGN KEY
MedicineID	INT	NOT NULL, FOREIGN KEY
Quantity	INT	NOT NULL, CHECK (> 0)
Price	DECIMAL(10,2)	NOT NULL, CHECK (> 0)

6 Customers Table

Attribute	Data Type	Constraints
CustomerID	INT	PRIMARY KEY, IDENTITY(1,1)
Name	VARCHAR(100)	NOT NULL
Phone	VARCHAR(20)	UNIQUE
Address	VARCHAR(255)	NULL

7 Sales Table

Attribute	Data Type	Constraints
SaleID	INT	PRIMARY KEY, IDENTITY(1,1)
CustomerID	INT	NOT NULL, FOREIGN KEY
SaleDate	DATE	NOT NULL, DEFAULT GETDATE()
TotalAmount	DECIMAL(10,2)	CHECK (>= 0)

8 SaleDetails Table (حل العلاقة Many-to-Many)

Attribute	Data Type	Constraints
SaleDetailID	INT	PRIMARY KEY, IDENTITY(1,1)
SaleID	INT	NOT NULL, FOREIGN KEY
MedicineID	INT	NOT NULL, FOREIGN KEY

Attribute	Data Type	Constraints
Quantity	INT	NOT NULL, CHECK (> 0)
Price	DECIMAL(10,2)	NOT NULL, CHECK (> 0)

9 Payments Table

Attribute	Data Type	Constraints
PaymentID	INT	PRIMARY KEY, IDENTITY(1,1)
SaleID	INT	NOT NULL, FOREIGN KEY
PaymentDate	DATE	DEFAULT GETDATE()
Amount	DECIMAL(10,2)	NOT NULL, CHECK (>= 0)
PaymentMethod	VARCHAR(50)	CHECK ('Cash', 'Card', 'Online')

10 Requests Table

Attribute	Data Type	Constraints
RequestID	INT	PRIMARY KEY, IDENTITY(1,1)
CustomerID	INT	NOT NULL, FOREIGN KEY
MedicineName	VARCHAR(100)	NOT NULL
RequestDate	DATE	DEFAULT GETDATE()
Status	VARCHAR(50)	DEFAULT 'Pending'

1 1 Complaints Table

Attribute	Data Type	Constraints
ComplaintID	INT	PRIMARY KEY, IDENTITY(1,1)
CustomerID	INT	NOT NULL, FOREIGN KEY
ComplaintText	VARCHAR(255)	NOT NULL
ComplaintDate	DATE	DEFAULT GETDATE()
Status	VARCHAR(50)	DEFAULT 'Open'
